

Binocular Rivalry

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ABSTRACT

Binocular rivalry is an example of how nonlinear dynamics occur in biology. Mathematical models of this phenomenon can be analyzed by using many important concepts and tools from the field of dynamics, such as linearization, limit cycles, and bifurcations. By varying the parameters in the model, the number of fixed points and their stabilities can change, resulting in very drastic differences in the results. Analyzing these results can then give a simple biological interpretation that explains what is occurring during the process of binocular rivalry. This paper will focus on analyzing a common model by applying these techniques from nonlinear dynamics, and then interpreting their meaning.

Keywords: Binocular Rivalry, Fixed Points Linearization, Limit Cycle, Hopf Bifurcation, Supercritical Pitchfork Bifurcation

INTRODUCTION

Binocular rivalry is a phenomenon that occurs when each eye is presented with a different image. The result is that the person perceives only one image at a time, determined by the firing rates of the neuron populations associated with each eye. Due to the tiring of neuron populations over time, this leads to an oscillatory behavior in the images the person perceives by the equations 1.1 - 1.4 (This model is based off of the one described by Rodica Curtu in *Proceedings in Applied Mathematics and Mechanics*) [1];

$$u_1' = -u_1 + S(I - \beta u_2 - g v_1) \quad (1.1)$$

$$u_2' = -u_2 + S(I - \beta u_1 - g v_2) \quad (1.2)$$

$$\tau v_1' = -v_1 + u_1 \quad (1.3)$$

$$\tau v_2' = -v_2 + u_2 \quad (1.4)$$

Where the constants β is the inhibitory interaction between the neural populations, g is the strength of adaptation (change in the neuron populations' response to the stimuli), τ is the adaptation time scale, and I is the strength of the stimuli (images that are very different have a greater I than images that are similar). The variables in this case represent the two neuron populations; the u variables represent the firing rates of the two neuron populations (a higher firing rate is equivalent to a greater activity level), and the v variables represents negative feedback adaptations such as synaptic depression (which can be thought of as a decrease in neurotransmitters, reducing activity levels)[2]. The function $S(x)$ is called the gain function and is normally represented as a version

of $S(x) = \frac{1}{1+e^{-x}} > 0$ (a simplified version of the gain function presented by Rodica Curtu)[1], and hence increases the rate of change of the firing rates, u' as seen in 1.1-1.2.

As we can see by equations 1.1-1.4, this is a 4-dimensional system with four parameters, so finding an exact solution would be nearly impossible. However, this system can still be analyzed by locating the fixed points and linearizing, as done in the following section. This will then allow us to analyze how the eigenvalues behave for different parameters, which can be used to explain what type of bifurcations are taking place. Finally, these results can illustrate what is happening biologically.

FIXED POINTS AND LINEARIZATION

The purpose for finding the fixed points and linearizing on them is to determine what type of fixed point they are, and more specifically their stability (which can then be used to find bifurcations). Figure 1 illustrates what type the fixed point is, based on the determinant and trace of the Jacobian. The eigenvalues determine the stability, but the determinant (product of eigenvalues) and trace (sum of eigenvalues) are based on the eigenvalues [3].

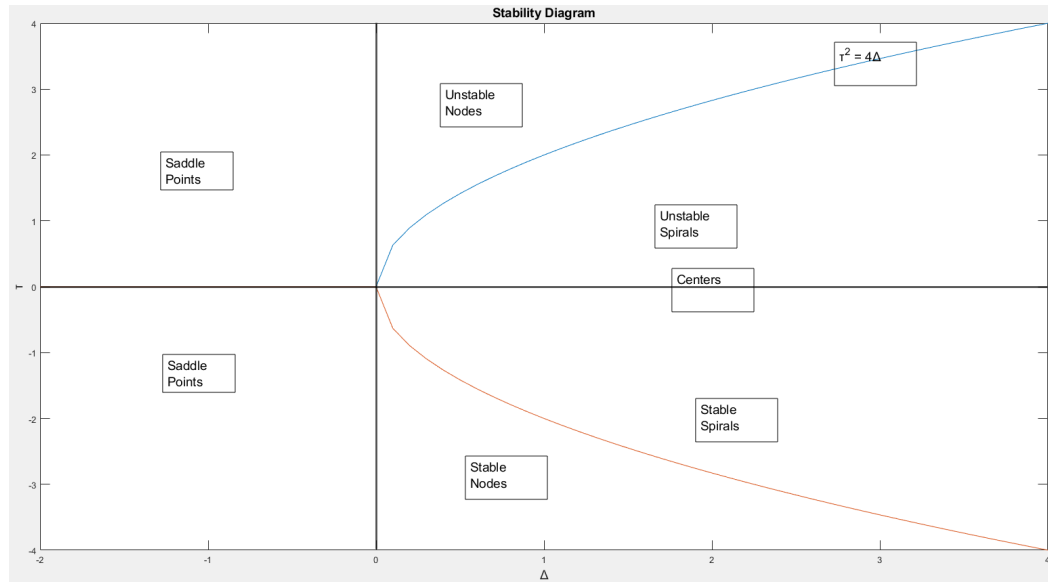


Figure 1. Stability Diagram

Fixed Point

A fixed point occurs when all the derivative terms are equal to zero, and the rate of change for each variable is zero. When setting equations 1.3 and 1.4 equal to zero, the only possible solutions are $u_1 = v_1$ and $u_2 = v_2$. Substituting this result into equations 1.1 and 1.2 and setting them equal to zero yields;

$$u'_1 = -u_1 + S(I - \beta u_2 - g u_1) = 0 \quad (1.5)$$

$$u'_2 = -u_2 + S(I - \beta u_1 - g u_2) = 0 \quad (1.6)$$

or equivalently that $u_1 = S(I - \beta u_2 - g u_1)$ and $u_2 = S(I - \beta u_1 - g u_2)$.

Next, let us assume that the activity levels are initially the same, so that $u_1 = u_2 = u_0$. Due to these equations being symmetric, it then yields $u_0 = S(I - \beta u_0 - g u_0)$. Furthermore, it can be shown that u_0 is unique by consider the two equations $y = x$ and $y = S(I - \beta x - g x)$. The first equation is increasing, and has a constant slope. The second equation is bounded between 0 and 1, and is always decreasing since the derivative is negative (with respect to x). As a result, these equations will only intersect one time, proving that u_0 will be a unique point [1].

Linearization

With the fixed point determined, it can now be linearized by computing the Jacobian matrix. For simplification, let $c = S'(x) = \frac{e^{-x}}{(1+e^{-x})^2} > 0$, evaluated at $x = I - \beta \mu_0 - g \mu_0$ [4]. Now letting the rows of the Jacobian correspond to 1.1, 1.3, 1.2, 1.4, respectively will give;

$$J = \begin{pmatrix} -1 & -gc & -\beta c & 0 \\ \frac{1}{\tau} & -\frac{1}{\tau} & 0 & 0 \\ -\beta c & 0 & -1 & -gc \\ 0 & 0 & \frac{1}{\tau} & -\frac{1}{\tau} \end{pmatrix} = \begin{pmatrix} A & B \\ B & A \end{pmatrix} \quad (1.7)$$

Notice that this matrix can be split into four blocks due to equivalence, to simplify the calculation of eigenvalues. Taking the determinant of each row element minus λI and setting it equal to zero yields a characteristic polynomial of the form;

$$\lambda^2 I - 2\lambda A + A^2 - B^2 = 0 \quad (1.8)$$

Using the quadratic equation then yields that the sets of eigenvalues are $\lambda_+ = A + B$ and $\lambda_- = A - B$, which each have two eigenvalues.

Eigenvalue Analysis

Recall that the trace is the sum of eigenvalues, and determinant is product of eigenvalues. For $A + B$, notice that the trace $T_+ = -1 - \frac{1}{\tau} - \beta c$ will be negative, whereas the determinant $\Delta_+ = \frac{1+gc+\beta c}{\tau}$ will be positive. As a result, the eigenvalues will always have a negative real part.

Now if we calculate the trace and determinant for $A - B$, we get the following results;

$$\Delta_- = \frac{c(g - \beta) + 1}{\tau} \quad (1.9)$$

$$T_- = \frac{-\tau - 1 + \beta c \tau}{\tau} \quad (1.10)$$

Since c is greater than one, the the determinant will be positive unless $\beta \gg g$ (the size of the difference will be based on how large or small c is), which will be analyzed later. Setting equation 1.10 to zero will show when the trace changes sign, which changes the fixed point between stable and unstable, or vice-versa, as seen in Figure 1. For trace greater than zero the point will be unstable, whereas for trace less than zero the point

will be stable. The following inequalities represent this (setting the numerator equal to zero and rearranging) [4];

$$c > \frac{1 + \frac{1}{\tau}}{\beta} \quad (1.11; T > 0)$$

$$c < \frac{1 + \frac{1}{\tau}}{\beta} \quad (1.12; T < 0)$$

The four parameters that can be varied are g, τ, β, I , where c is dependent on I, μ_0, g, β . This then results in various cases to analyze, which can result in different phenomenon depending on which variable is treated as a parameter that can be varied and the others treated as constants. The fixed point u_0 can change stability by either the real parts of the third and fourth eigenvalue crossing the imaginary axis (they change from negative to positive), meaning the trace will go from negative to positive, or by one eigenvalue equaling to zero while the other eigenvalue is less than zero [5]. This means the determinant would achieve zero while the trace is still negative. If determinant 1.9 is always positive, then this implies the case where an eigenvalue will never be zero. To demonstrate the case where an eigenvalue crosses zero, then both 1.10 and 1.9 can be set to zero, and an inequality can be established so that 1.9 will reach zero while 1.10 is still negative.

BIFURCATION ANALYSIS

First the bifurcations associated with the trace changing signs will be analyzed by choosing appropriate parameters. Secondly, the bifurcations associated with the determinant reaching zero first will be analyzed by determining what parameters lead to this occurrence.

Positive Determinant

Letting $g > \beta$, the determinant will be positive. For a large τ , the trace will approximately be (from equations 1.11 - 1.12);

$$\beta c \approx 1 \quad (1.13)$$

Writing c out and expanding this equation yields;

$$\beta e^{-x} \approx e^{-2x} + 2e^{-x} + 1 \implies e^{-2x} + (2 - \beta)e^{-x} + 1 \approx 0$$

Letting $y = e^{-x}$ and using the quadratic equation;

$$y \approx \frac{\beta - 2 \pm \sqrt{(2 - \beta)^2 - 4}}{2} \quad (1.14)$$

Since y must be positive, we see that this will have two solutions for a sufficiently large β , suggesting that the trace can change signs two times. This indicates that there are two c_1, c_2 corresponding to I_1, I_2 where the sign changes from negative to positive, then back to negative. However, this also shows that for a small enough β , such that $|2 - \beta| < 2$, then equation 1.13 cannot be satisfied, and the trace will not change signs. this will

result in the trace never becoming positive, and so by figure 1 the fixed point μ_0 will remain stable.

Going back to the case where y has two solutions; for $c < c_1$ the trace is negative, representing a stable spiral by Figure 1. Once $c_2 > c > c_1$, then the fixed point will become an unstable spiral by Figure 1. Therefore, this suggests that a Hopf Bifurcation is occurring when $I = I_1, c = c_1$. Furthermore, for $c > c_2$ the trace will become negative again, going back to a stable spiral. This indicates that another Hopf bifurcation is occurring at $c = c_2, I = I_2$, so that there will be two bifurcation points [1]. Since the phase portrait would be in 4D and nearly impossible to visualize, we will instead observe the time series for the variables, focusing on u_1 and u_2 since they correspond to the neuron activity levels. Letting $\tau = 100, g = 7.6, \beta = 7.5$, the time series can be plotted for various values of I less than I_1 , between I_1, I_2 , and greater than I_2 .

Figure 2 shows the time series for $I = .001$, indicating that there is a fixed point since all of the initial values for u_1 and u_2 converge to the same point. Figure 3 shows the time series for $I = 1$, now indicating that $I_2 > I > I_1$, and that the first bifurcation has occurred. This is a supercritical Hopf bifurcation because we can see that both u_1 and u_2 converge to a limit cycle by the periodicity exhibited, which means that it is stable.

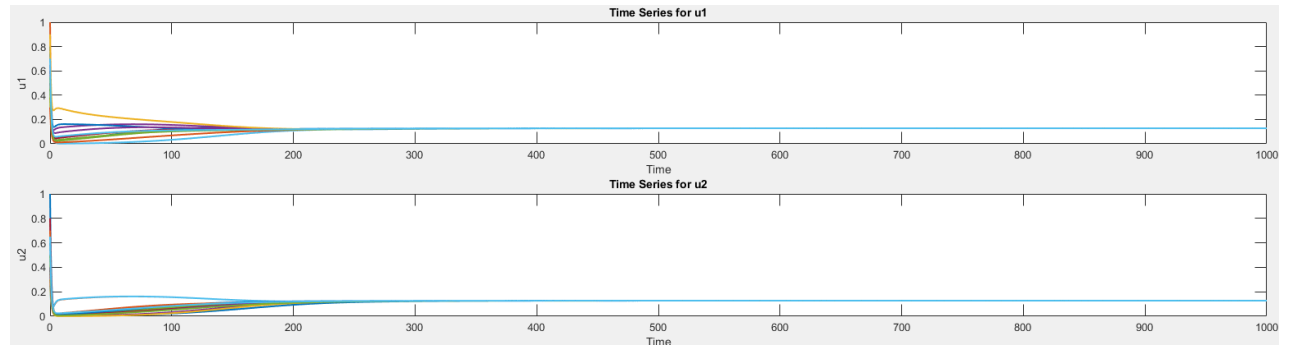


Figure 2. Time Series of μ_1 and μ_2 before the bifurcation, $I = .001$

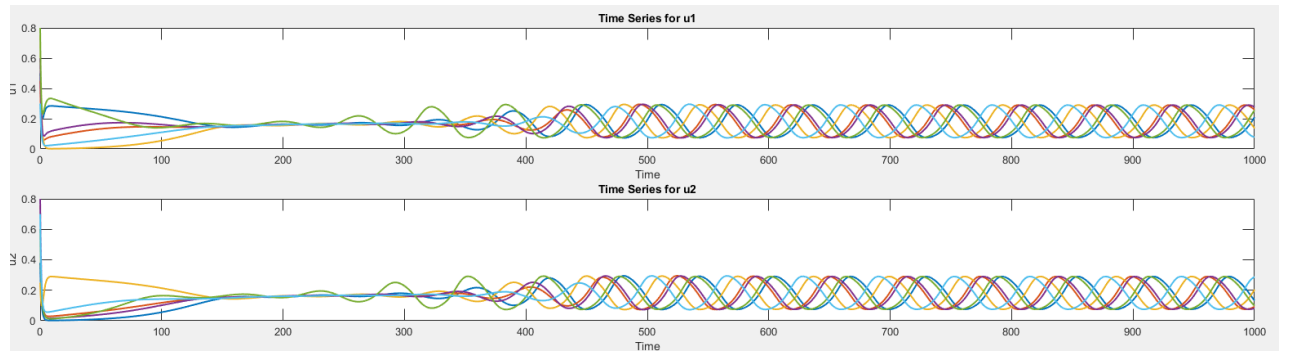


Figure 3. Time Series of μ_1 and μ_2 after bifurcation, $I = 1$

Figure 4 shows the time series of one trajectory for an intensity greater than the first bifurcation, at $I = 5$, but still less than I_2 . This shows that the limit cycle has grown much larger and the period has also increased, which is expected from a supercritical Hopf Bifurcation. Furthermore, this figure shows that while u_1 is at its peak, u_2 is at its

minimum, and vice-versa. However, as $I \rightarrow I_2$, this limit cycle is expected to become smaller until converging to a fixed point.

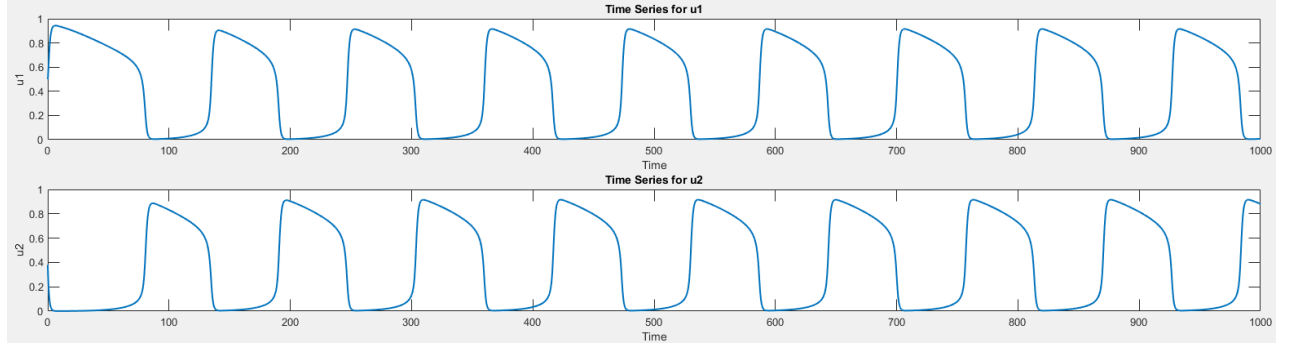


Figure 4. Time Series of u_1 and u_2 for one trajectory after the bifurcation, $I = 5$

Now Figure 5 shows the time series of multiple initial conditions for $I = 15 > I_2$, confirming that the fixed point is a stable spiral and showing that the limit cycle has converged. This change from a stable limit cycle to a stable spiral is again a supercritical Hopf Bifurcation.

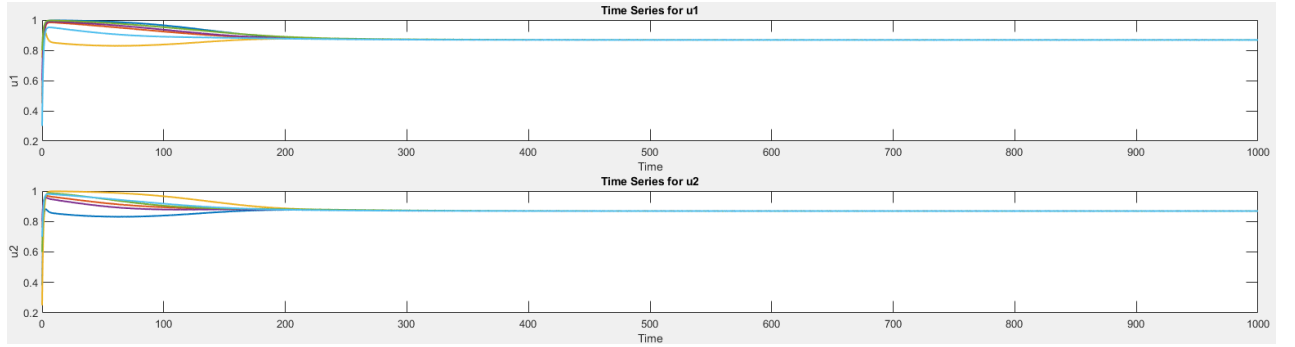


Figure 5. Time Series of u_1 and u_2 after the second bifurcation, $I = 15$

Zero Eigenvalue

The change in stability due to an eigenvalue being zero (resulting from the determinant changing signs) occurs when equation 1.9 is equal to zero, and equation 1.10 is negative (otherwise, the previous case would also occur due to a change in the sign of the trace). Setting the determinant 1.9 to zero yields;

$$c = \frac{1}{\beta - g} \quad (1.15)$$

Setting the trace 1.10 less than or equal to zero yields the inequality;

$$g(\tau + 1) > \beta \quad (1.16)$$

As with before, we can take $\tau = 100$, which changes this inequality to $101g > \beta$ and since c cannot be negative, $\beta > g$. As with our analysis with the Hopf Bifurcation, we can expand c out and write $y = e^{-x}$ for equation 1.15 to get

$$(\beta - g)y = y^2 + 2y + 1 \implies y^2 + (2 + g - \beta)y + 1 = 0$$

Using the quadratic equation;

$$y = \frac{\beta - 2 - g \pm \sqrt{(2 + g - \beta)^2 - 4}}{2} \quad (1.17)$$

Again y is nonnegative, so for two solutions to occur β must be significantly larger than g (so that both the plus and minus cases will fall into the range of y). If $|2 + g - \beta| < 2$, then there will be no solutions to this equation, and a bifurcation will not occur. The determinant will remain positive, which by figure 1 suggests that this fixed point will remain stable.

To ensure the case where two solutions occur, we can take $g = .1$ and $\beta = 5$. As with before, we can let these points be c_1, c_2 corresponding to I_1, I_2 [5]. Figure 6 plots the 2 dimensional phase portrait (for u_1, u_2), with $I = .001 < I_1$, showing that there exists a stable fixed point.

Figure 7 plots the 2 dimensional phase portrait (for u_1, u_2), with $I = 1 > I_1, 1 < I_2$. This shows that there are three fixed points, 2 of which are stable, and one which is unstable. The two stable ones are to the upper left and bottom right, respectively, while the unstable one lies between these two, occurring for initial conditions when $u_1 = u_2$. Going from one stable fixed point to two stable and one unstable indicates a supercritical pitchfork bifurcation.

Figure 8 plots the 2 dimensional phase portrait (for u_1, u_2), with $I = 5 > I_2$. This shows that there is one stable fixed point, indicating that the bifurcation occurring at I_2 is again a supercritical pitchfork bifurcation.

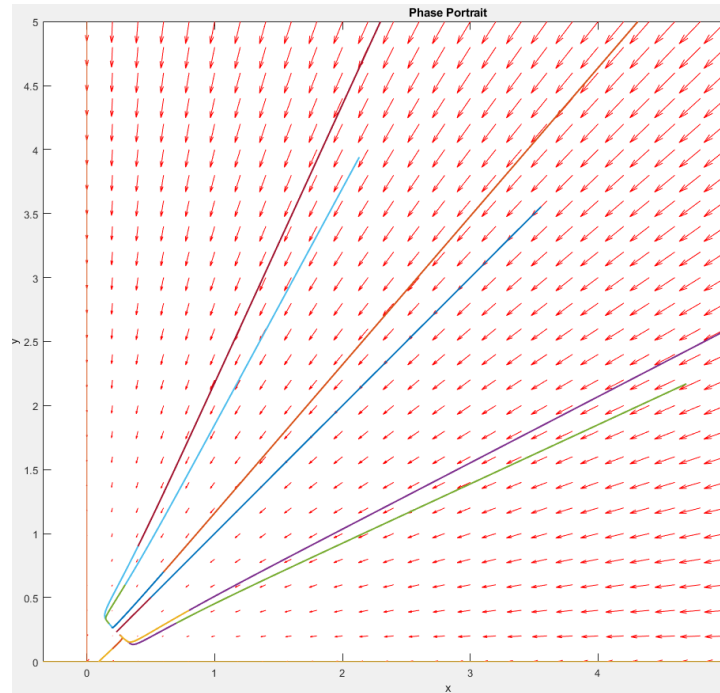


Figure 6. Phase Portrait of u_1 and u_2 before first bifurcation, $I = .001$

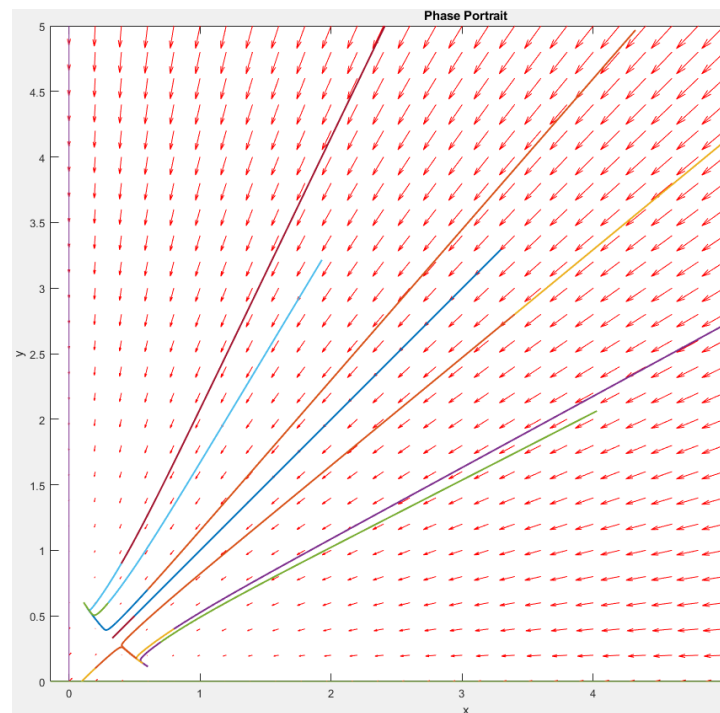


Figure 7. Phase Portrait of u_1 and u_2 after first bifurcation, $I = 1$

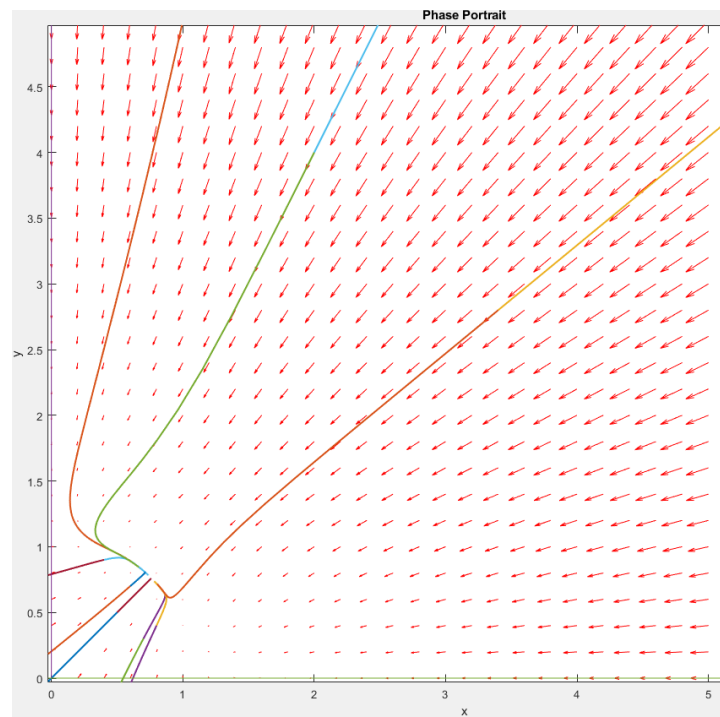


Figure 8. Phase Portrait of u_1 and u_2 after second bifurcation, $I = 5$

BIOLOGICAL INTERPRETATION

Now that the bifurcations for these two cases have been analyzed, the biological results of each case can be discussed. First the case where a supercritical Hopf bifurcation (positive determinant) occurs will be analyzed, and secondly the case where the supercritical pitchfork bifurcation (zero eigenvalue) occurs will be analyzed.

Supercritical Hopf Bifurcation

This result has the parameters $g > \beta$, meaning that the inhibitory interaction β will be less than the strength of adaptation, with the time scale very large. The intensity I of the contrast between images determines what the person will perceive. If β is not sufficiently large enough, then regardless of the intensity, the two firing rates will converge to a fixed point and neither will dominate.

On the other hand, if β is large enough, the bifurcation will occur. For an intensity less than the critical one, the contrast between the images is not great enough. As a result, both neuron populations converge to a fixed point and neither of the images dominate. However, once the intensity increases beyond the first critical point the contrast in the images is great enough so that a limit cycle will occur. That is, one population will initially dominate, but then inhibitory reactions along with adaptation causes its firing rate to decrease. As a result, the other neuron population's firing rate increases and then dominates. The person will then see an alternating pattern of the two images, and as the intensity increases the period of this alternation grows. Once the intensity grows too large, the limit cycles will then begin to shrink, and the firing rates u_1 and u_2 will converge to a fixed point again, where neither neuron population dominates.

Supercritical Pitchfork Bifurcation

This result has the parameters g small and $\beta \gg g$, meaning that the strength of adaptation is very minimal, while the time scale τ is very large. If β is not large enough, or g is too large, there will not be a bifurcation. Therefore, regardless of the intensity, the firing rates will again converge to a fixed point where neither neuron population dominates.

However, if β is large enough and g is also small enough, then the bifurcation will occur. For a small intensity I , the firing rates u_1 and u_2 for both populations will converge to a fixed point, with neither of the images dominating. After the bifurcation, for $u_1 \neq u_2$ initially, they will converge to either a fixed point where $u_1 > u_2$, or $u_1 < u_2$. This results in a winner-take-all situation, where the image seen by the viewer is the one associated with a higher firing rate [6]. Once the intensity I has become large enough, the firing rates again converge to a fixed point, where neither image dominates the other.

SUMMARY

This paper analyzed the dynamics of a simple model for binocular rivalry, focusing on different possible types of bifurcations by varying the intensity parameter, while keeping the other parameters fixed. The first step was locating a fixed point, and then linearizing around this point. This then allowed the eigenvalues of the linearized system to be studied, which tell what type of stability change occurs to the fixed point, if any change occurs. One possibility analyzed was the the supercritical Hopf, which occurred for large β where $g > \beta$. This resulted in a stable limit cycle between two bifurcation

points, I_1, I_2 , while for intensities outside of this range there was a single stable fixed point. The other possibility analyzed was a supercritical pitchfork, which occurred for large β and small g . This resulted in two stable and one unstable fixed points between I_1, I_2 , while for other intensities falling outside of this range resulted in one stable fixed point.

Due to the complexity of this system, choosing different initial values and fixing different combinations of parameters while varying a single parameter can result in the discovery of other possible bifurcations. This can then lead to further research on binocular rivalry by suggesting different experiments to conduct on various parameters, and observing if they match results predicted from the model. Furthermore, while this is only a model and so cannot describe binocular rivalry exactly, it does provide a deeper understanding for how this process works and is able to explain the results of experiments conducted.

ACKNOWLEDGMENTS

I would like to thank professor Bard Ermentrout, currently at the University of Pittsburgh, for his recommendation of several resources used in this paper. The textbook Nonlinear Dynamics and Chaos by Strogatz was also a major source of inspiration for this project [7]. As for the figures used, all of them were created using MATLAB.

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[1] [2] [3] [4] [5] [6] [7]

MATLAB CODE

FIGURE 1:

```
x = -2:.1:4;  
y = 2.*sqrt(x);  
plot(x,y)  
hold on;  
  
y = -2.*sqrt(x);  
plot(x,y)  
hold on;  
  
xlabel('Δ');  
ylabel('τ');  
  
xline(0, 'Color', 'k', 'LineWidth', 2);  
yline(0, 'Color', 'k', 'LineWidth', 2);
```

FIGURE 2:

```
f = @(x, y, w, z) -x + 1./(1 + exp(-1*(.001 - 7.5.*y - 7.6.*w)));
g = @(x, y, w, z) -y + 1./(1 + exp(-1*(.001 - 7.5.*x - 7.6.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x, y, w, z) (-z + y)./100;
```

% Define the ODE system

```
system = @(t, Y) [
    f(Y(1), Y(2), Y(3), Y(4)); % dx/dt
    g(Y(1), Y(2), Y(3), Y(4)); % dy/dt
    h(Y(1), Y(2), Y(3), Y(4)); % dw/dt
    k(Y(1), Y(2), Y(3), Y(4)); % dz/dt
];
```

```
initial_conditions = [
    0.5 1.0 0.5 1.0;
    1.0 0.5 0.75 0.25;
    0.38 0.5 0.38 0.5;
    0.5 0.38 0.25 0.75;
    0.45 0.76 0.45 0.55;
    0.75 0.25 0.75 0.25;
    0.6 0.8 0.4 0.6;
    0.8 0.6 0.2 0.4;
    0.3 0.7 0.6 0.5;
    0.9 0.3 0.1 0.8;
    0.65 0.4 0.3 0.6;
    0.55 0.5 0.5 0.7;
    0.7 0.65 0.35 0.45;
];
```

```
figure;
subplot(4, 1, 1);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 1), 'LineWidth', 1.5); % x vs time
    hold on;
end
xlabel('Time');
ylabel('u1');
title('Time Series for u1');

subplot(4, 1, 2);
```

```

for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 2), 'LineWidth', 1.5); % y vs time
    hold on;
end
xlabel('Time');
ylabel('u2');
title('Time Series for u2');

subplot(4, 1, 3);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 3), 'LineWidth', 1.5); % w vs time
    hold on;
end
xlabel('Time');
ylabel('v1');
title('Time Series for v1');

subplot(4, 1, 4);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 4), 'LineWidth', 1.5); % z vs time
    hold on;
end
xlabel('Time');
ylabel('v2');
title('Time Series for v2');

```

FIGURE 3:

```
f = @(x, y, w, z) -x + 1./(1 + exp(-1*(1 - 7.5.*y - 7.6.*w)));
g = @(x, y, w, z) -y + 1./(1 + exp(-1*(1 - 7.5.*x - 7.6.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x, y, w, z) (-z + y)./100;

% Define the ODE system
system = @(t, Y) [
    f(Y(1), Y(2), Y(3), Y(4)); % dx/dt
    g(Y(1), Y(2), Y(3), Y(4)); % dy/dt
    h(Y(1), Y(2), Y(3), Y(4)); % dw/dt
    k(Y(1), Y(2), Y(3), Y(4)); % dz/dt
];

initial_conditions = [
    0.5 0.38 0.25 0.75;
    0.45 0.76 0.45 0.55;
    0.75 0.25 0.75 0.25;
    0.6 0.8 0.4 0.6;
    0.8 0.6 0.2 0.4;
    0.3 0.7 0.6 0.5;
];

figure;
subplot(4, 1, 1);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 1), 'LineWidth', 1.5); % x vs time
    hold on;
end
xlabel('Time');
ylabel('u1');
title('Time Series for u1');

subplot(4, 1, 2);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 2), 'LineWidth', 1.5); % y vs time
    hold on;
end
xlabel('Time');
ylabel('u2');
```

```

title('Time Series for u2');

subplot(4, 1, 3);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 3), 'LineWidth', 1.5); % w vs time
    hold on;
end
xlabel('Time');
ylabel('v1');
title('Time Series for v1');

subplot(4, 1, 4);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 4), 'LineWidth', 1.5); % z vs time
    hold on;
end
xlabel('Time');
ylabel('v2');
title('Time Series for v2');

```

FIGURE 4:

```
f = @(x, y, w, z) -x + 1./(1 + exp(-1.*(5 - 7.5.*y - 7.6.*w)));
g = @(x, y, w, z) -y + 1./(1 + exp(-1.*(5 - 7.5.*x - 7.6.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x, y, w, z) (-z + y)./100;

% Define the ODE system
system = @(t, Y) [
    f(Y(1), Y(2), Y(3), Y(4)); % dx/dt
    g(Y(1), Y(2), Y(3), Y(4)); % dy/dt
    h(Y(1), Y(2), Y(3), Y(4)); % dw/dt
    k(Y(1), Y(2), Y(3), Y(4)); % dz/dt
];

initial_conditions = [
    0.5 0.38 0.25 0.75;
];

time_horizon = [0, 1000];

figure;
subplot(4, 1, 1);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 1), 'LineWidth', 1.5); % x vs time
    hold on;
end
xlabel('Time');
ylabel('u1');
title('Time Series for u1');

subplot(4, 1, 2);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 2), 'LineWidth', 1.5); % y vs time
    hold on;
end
xlabel('Time');
ylabel('u2');
title('Time Series for u2');

subplot(4, 1, 3);
```



```

for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 3), 'LineWidth', 1.5); % w vs time
    hold on;
end
xlabel('Time');
ylabel('v1');
title('Time Series for v1');

subplot(4, 1, 4);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 4), 'LineWidth', 1.5); % z vs time
    hold on;
end
xlabel('Time');
ylabel('v2');
title('Time Series for v2');

```

FIGURE 5:

```
f = @(x, y, w, z) -x + 1./(1 + exp(-1*(15 - 7.5.*y - 7.6.*w)));
g = @(x, y, w, z) -y + 1./(1 + exp(-1*(15 - 7.5.*x - 7.6.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x, y, w, z) (-z + y)./100;

system = @(t, Y) [
    f(Y(1), Y(2), Y(3), Y(4)); % dx/dt
    g(Y(1), Y(2), Y(3), Y(4)); % dy/dt
    h(Y(1), Y(2), Y(3), Y(4)); % dw/dt
    k(Y(1), Y(2), Y(3), Y(4)); % dz/dt
];

initial_conditions = [
    0.5 0.38 0.25 0.75;
    0.45 0.76 0.45 0.55;
    0.75 0.25 0.75 0.25;
    0.6 0.8 0.4 0.6;
    0.8 0.6 0.2 0.4;
    0.3 0.7 0.6 0.5;
];

figure;
subplot(4, 1, 1);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 1), 'LineWidth', 1.5); % x vs time
    hold on;
end
xlabel('Time');
ylabel('u1');
title('Time Series for u1');

subplot(4, 1, 2);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 2), 'LineWidth', 1.5); % y vs time
    hold on;
end
xlabel('Time');
ylabel('u2');
title('Time Series for u2');
```

```

subplot(4, 1, 3);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 3), 'LineWidth', 1.5); % w vs time
    hold on;
end
xlabel('Time');
ylabel('v1');
title('Time Series for v1');

subplot(4, 1, 4);
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, time_horizon, initial_conditions(i, :));
    plot(t, sol(:, 4), 'LineWidth', 1.5); % z vs time
    hold on;
end
xlabel('Time');
ylabel('v2');
title('Time Series for v2');

```

FIGURE 6:

```
f = @(x,y,w,z) -x + 1./(1 + exp(-1.*(0.001 - 5.*y - .1.*w)));
g = @(x,y,w,z) -y + 1./(1 + exp(-1.*(0.001 - 5.*x - .1.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x,y,w,z) (-z + y)./100;
```

```
system = @(t, Y)[
    f(Y(1), Y(2), Y(3), Y(4));
    g(Y(1), Y(2), Y(3), Y(4));
    h(Y(1), Y(2), Y(3), Y(4));
    k(Y(1), Y(2), Y(3), Y(4));
];
```

```
initial_conditions = [
    0.2 0.1 0.3 0.4;
    0.7 0.3 0.1 0.6;
    0.4 0.9 0.2 0.5;
    0.6 0.7 0.8 0.2;
    0.8 0.4 0.5 0.3;
    0.3 0.6 0.7 0.1;
    0.5 0.5 0.5 0.5;
];
```

```
[x, y] = meshgrid(0:.2:5, 0:.2:5);
z = .25;
w = .35;
u = f(x,y,w,z);
v = g(x,y,w,z);
```

```
figure;
quiver(x, y, u, v, 'r');
title('Phase Portrait');
axis([0 5 0 5]);
axis equal;
hold on;
```

```
for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, [0,10], initial_conditions(i,:));
    plot(sol(:,1), sol(:,2), 'LineWidth', 1.5);
```

```
[t, sol] = ode45(system, [0,-2], initial_conditions(i,:));
```

```
    plot(sol(:,1), sol(:,2), 'LineWidth', 1.5);  
end;
```

```
xlabel('x');  
ylabel('y');  
xL = xlim;  
yL = ylim;  
line([0 0], yL);  
line(xL, [0 0]);
```

```
hold off;
```

FIGURE 7:

```
f = @(x,y,w,z) -x + 1./(1 + exp(-1.*(1 - 5.*y - .1.*w)));
g = @(x,y,w,z) -y + 1./(1 + exp(-1.*(1 - 5.*x - .1.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x,y,w,z) (-z + y)./100;

system = @(t, Y)[
    f(Y(1), Y(2), Y(3), Y(4));
    g(Y(1), Y(2), Y(3), Y(4));
    h(Y(1), Y(2), Y(3), Y(4));
    k(Y(1), Y(2), Y(3), Y(4));
];

initial_conditions = [
    0.2 0.1 0.3 0.4;
    0.7 0.3 0.1 0.6;
    0.4 0.9 0.2 0.5;
    0.6 0.7 0.8 0.2;
    0.8 0.4 0.5 0.3;
    0.3 0.6 0.7 0.1;
    0.5 0.5 0.5 0.5;
    3.4 2.8 0.1 0.2;
];

[x, y] = meshgrid(0:.2:5, 0:.2:5);
z = .25;
w = .35;
u = f(x,y,w,z);
v = g(x,y,w,z);

figure;
quiver(x, y, u, v, 'r');
title('Phase Portrait');
axis([0 5 0 5]);
axis equal;
hold on;

for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, [0,15], initial_conditions(i,:));
    plot(sol(:,1), sol(:,2), 'LineWidth', 1.5);
```

```
[t, sol] = ode45(system, [0,-2], initial_conditions(i,:));  
plot(sol(:,1), sol(:,2), 'LineWidth', 1.5);  
end;
```

```
xlabel('x');  
ylabel('y');  
xL = xlim;  
yL = ylim;  
line([0 0], yL);  
line(xL, [0 0]);
```

```
hold off;
```

FIGURE 8:

```
f = @(x,y,w,z) -x + 1./(1 + exp(-1.*(5 - 5.*y - .1.*w)));
g = @(x,y,w,z) -y + 1./(1 + exp(-1.*(5 - 5.*x - .1.*z)));
h = @(x, y, w, z) (-w + x)./100;
k = @(x,y,w,z) (-z + y)./100;

system = @(t, Y)[
    f(Y(1), Y(2), Y(3), Y(4));
    g(Y(1), Y(2), Y(3), Y(4));
    h(Y(1), Y(2), Y(3), Y(4));
    k(Y(1), Y(2), Y(3), Y(4));
];

initial_conditions = [
    0.2 3 0.3 0.4;
    0.7 0.3 0.1 0.6;
    0.6 0.7 0.8 0.2;
    0.8 0.4 0.5 0.3;
    1.0 2.6 0.7 0.1;
    0.5 0.5 0.5 0.5;
    3.4 2.8 0.1 0.2;
];

[x, y] = meshgrid(0:.2:5, 0:.2:5);
z = .25;
w = .35;
u = f(x,y,w,z);
v = g(x,y,w,z);

figure;
quiver(x, y, u, v, 'r');
title('Phase Portrait');
axis([0 5 0 5]);
axis equal;
hold on;

for i = 1:size(initial_conditions, 1)
    [t, sol] = ode45(system, [0,15], initial_conditions(i,:));
    plot(sol(:,1), sol(:,2), 'LineWidth', 1.5);

    [t, sol] = ode45(system, [0,-2], initial_conditions(i,:));
```



```
    plot(sol(:,1), sol(:,2), 'LineWidth', 1.5);  
end;
```

```
xlabel('x');  
ylabel('y');  
xL = xlim;  
yL = ylim;  
line([0 0], yL);  
line(xL, [0 0]);
```

```
hold off;
```